

Trainings ▾

Preparatory Steps

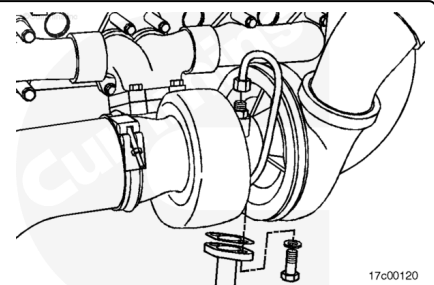
ISX Automotive with CM570, QSX15 with CM570, and Power Generation with CM570

- Remove the wastegate actuator hose. Refer to Procedure 010-050 in Section 10.
([/qs3/pubsys2/xml/en/procedures/10/10-010-050.html](http://qs3/pubsys2/xml/en/procedures/10/10-010-050.html))

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

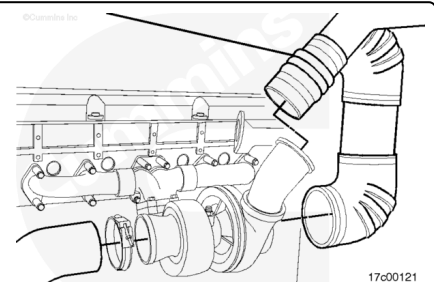


Automotive, Industrial, and Power Generation with CM570

- Remove the oil supply, oil drain tube, and brace, if equipped, from the turbocharger.

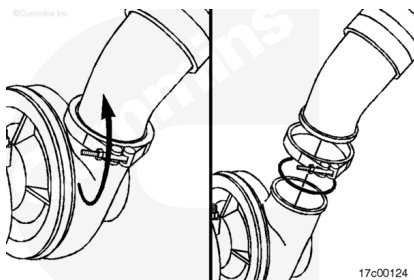
Remove the intake and exhaust pipes from the turbocharger.

Remove the charge air cooler piping from the discharge elbow.



Remove the clamp, o-ring, and discharge elbow from the turbocharger.

Discard the o-ring.

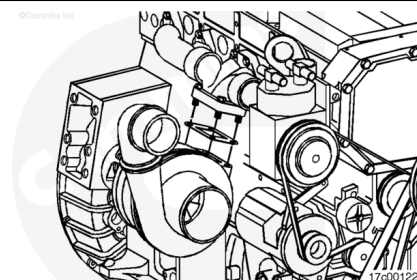


Note : In some applications, the turbocharger will **not** clear the lubricating oil cooler assembly during removal and installation. It will, perhaps, be necessary to remove the exhaust manifold and turbocharger together, and then separate the two components. Use the following procedure if the two components must be removed together for the removal and installation of the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html>)

If the turbocharger mounting nuts do **not** loosen freely, split the nuts to avoid breaking a mounting stud.

Remove the four turbocharger mounting nuts.

Remove the turbocharger and discard the gasket.



Install

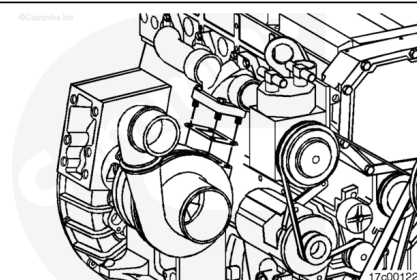
ISX Automotive with CM570, QSX15 with CM570, and Power Generation with CM570

- Apply a film of high-temperature anti-seize compound, Part Number 3823097, or equivalent, to the turbocharger mounting studs.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : If the exhaust manifold and turbocharger were removed together in order for the turbocharger to clear the lubricating oil cooler assembly, see the following procedure for installation of the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html>)

Install a new mounting gasket, the turbocharger, and the four mounting nuts.

Tighten the mounting nuts.

Torque value for the standard nut:

Torque Value: 61 n•m [45 ft-lb]

Torque value for the Spirallock™ nut (identified by "SPL" on the nut flange):

Torque Value: 81 n•m [60 ft-lb]

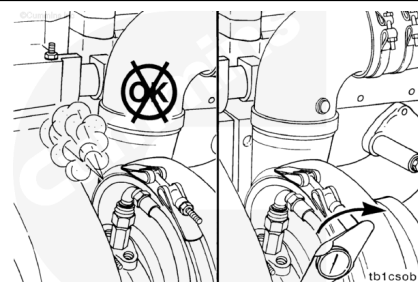
If installing a new turbocharger, it can be necessary to rotate the turbocharger compressor housing to properly align with the charge air cooler piping.

Rotate the turbocharger compressor housing by loosening the V-band between the turbocharger bearing housing and the turbocharger compressor housing.

Rotate the turbocharger compressor housing to properly align with the charge air cooler piping.

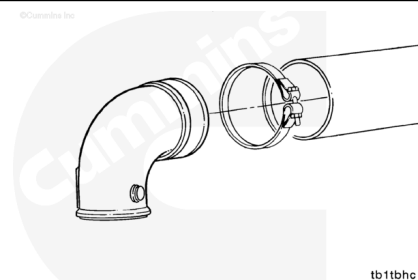
Tighten the V-band clamp.

Torque Value: 8 n•m [71 in-lb]



Install the discharge elbow and clamp onto the charge air cooler pipe connection.

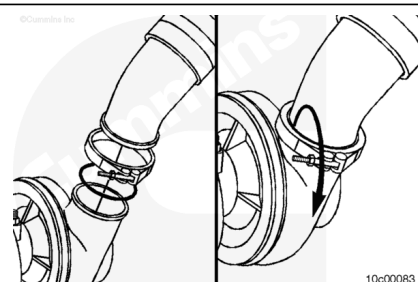
Do **not** tighten the clamp until the elbow is installed on the turbocharger.



If a new turbocharger is being installed, the variable geometry turbocharger actuator from the original turbocharger **must** be reused.

Refer to Procedure 010-134 in Section 10.

(</qs3/pubsys2/xml/en/procedures/10/10-010-134.html>)



Install a new o-ring, clamp, and discharge elbow to the turbocharger.

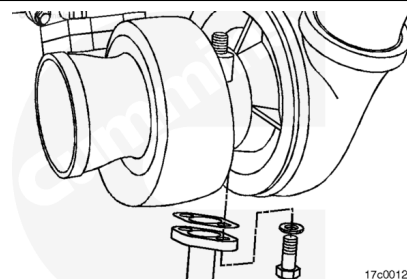
Tighten the clamp.

Torque Value: 9 n•m [80 in-lb]

Install a new gasket, oil drain tube, and capscrews.

Tighten the capscrews.

Torque Value: 25 n•m [221 in-lb]

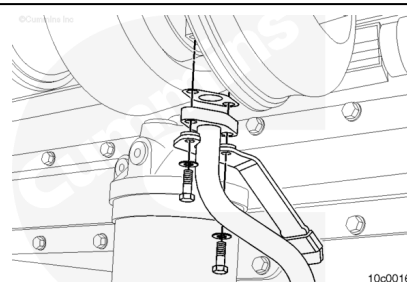


Install the turbocharger anti-rotation brace, if equipped. The end rests on the oil cooler housing.

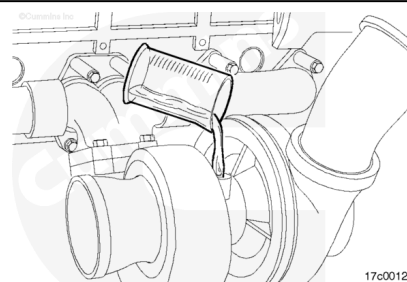
This brace requires longer, 1.25 mm [0.049 in], capscrews.

Tighten the capscrews.

Torque Value: 25 n•m [221 in-lb]

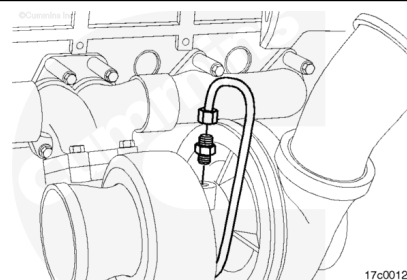


Pour approximately 50 to 60 cc [1.7 to 2 oz] of clean engine oil into the turbocharger oil supply opening.



⚠ CAUTION ⚠

Proper routing of the turbocharger oil supply tube is critical to prevent failure. Avoid any tube-to-metal contact. (The inlet supply fitting must be oriented slightly off vertical to allow proper alignment.)



If installing a new turbocharger, make sure the turbocharger is aligned, loosen the compressor and turbine V-bands, and adjust as needed. Tighten the V-bands.

Torque Value: 9 n•m [80 in-lb]

Torque Value: 12 n•m [106 in-lb]

If installing a new turbocharger, install the male union elbow.

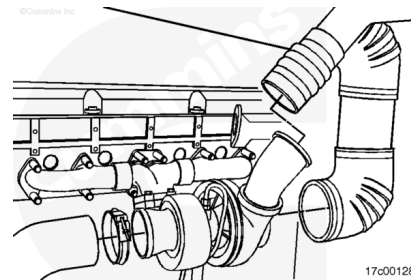
Torque Value: 30 N•m [22 ft-lb]

Install the turbocharger oil supply tube on the elbow.

Torque Value: 30 N•m [22 ft-lb]

Install the intake and exhaust pipes to the turbocharger.
Tighten the clamp.

Torque Value: 9 N•m [80 in-lb]



Finishing Steps

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- Install the wastegate actuator hose. Refer to Procedure 010-050 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-050.html>)
- Start and operate the engine until the coolant temperature reaches 82°C [180°F]. Check for air, coolant, and oil leaks.